

Security+ 2.4 · An Overview of Malware

Mestre CyberSec · Lesson 2.4 — An Overview of Malware (6:06) · [Assistir Vídeo no YouTube ↗](#) · Treino Interativo: edusouzamarques.github.io/cybersec

Executive Lesson Summary: Malware = software malicioso. Categoriza: INTENÇÃO (spionagem, roubo, derrubar — DoS); FORMA (virus/worm/rootkit/trojan/keylogger/ransomware/stealer); ESCONDER (packer/compress/packing, polymorphic, metamorphic, fileless).

💡 Key Exam Highlights & Critical Concepts

- Fileless = roda na memória, não toca em disco — invisível pro AV
- Polymorphic/métamorphic muda de forma — evita assinatura

🎯 Exam Tips & Typical CompTIA Scenarios

- 'malware without a file on disk' = fileless malware

Beyond the Video: Official Exam Concepts, Protocols & Traps (Section 2.4 — An Overview of Malware)

Why is this advanced coverage required for SY0-701? Video lectures provide foundational concepts, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

• Virus (N/A)

Malicious code that attaches to a legitimate host file or program and requires user execution or file sharing to spread.

 **Exam Trap:** *Precisa de um hospedeiro e de ação do usuário; não se auto-replica como um worm.*

• Worm (N/A)

Self-replicating malware that spreads across networks without requiring a host file or user interaction.

 **Exam Trap:** *Auto-replica e se propaga sozinho pela rede; a ausência de interação do usuário é a chave.*

• Trojan Horse (N/A)

Malware disguised as legitimate software, which users are tricked into installing; it does not self-replicate.

 **Exam Trap:** *Disfarce de programa legítimo; não se propaga sozinho como vírus ou worm.*

• Ransomware (N/A)

Malware that encrypts files or locks systems and demands payment for the decryption key or restoration.

 **Exam Trap:** *Criptografia de dados + resgate; backups offline são a principal mitigação.*

• Keylogger (N/A)

Malware or hardware that records keystrokes to capture credentials, personal data, or other sensitive information.

 **Exam Trap:** *Registra teclas; pode ser software ou hardware; associado a roubo de credenciais.*

• Rootkit (N/A)

Malware designed to gain privileged access and hide its presence and other malicious activities from the OS and security tools.

 **Exam Trap:** *Oculto a si mesmo e outras ameaças; detecção é difícil, muitas vezes exige boot externo ou ferramenta especializada.*

• Fileless Malware (N/A)

Malware that executes in memory using legitimate system tools or scripts without writing a traditional file to disk.

 **Exam Trap:** *Gatilho: 'malware without a file on disk' = fileless; difícil para antivírus baseado em arquivo.*

• Polymorphic/Metamorphic Malware (N/A)

Malware that changes its code or appearance to evade signature-based detection; polymorphic mutates but keeps core payload, metamorphic rewrites its entire code.

 **Exam Trap:** *Muda de forma para evitar assinaturas; polimórfico usa criptografia/mutação, metamórfico reescreve o código.*

Authentic Practice Questions (10 Real Exam Scenarios)

Scenario-based exam questions. Complete explanations and distractor analysis are provided in the answer key below.

QUESTION 1 · EXAM SCENARIO

An accounting employee opens an email attachment and enables macros when prompted. Days later, the security team finds that executable files on a shared department drive have been modified and that other users' workstations become infected only after they run those modified files. The malicious code does not spread by itself across the network. Which malware type best matches this behavior?

- A) Worm
- B) Virus
- C) Rootkit
- D) Ransomware

QUESTION 2 · EXAM SCENARIO

After a laptop is connected to a hospital's internal network, multiple unpatched workstations begin showing SMB connection attempts and become compromised within minutes. No email attachment or user download is involved, and the infection spreads continuously from host to host. Which malware type is most likely involved?

- A) Virus
- B) Worm
- C) Trojan horse
- D) Keylogger

QUESTION 3 · EXAM SCENARIO

An employee downloads a utility from a third-party website because it promises faster PDF conversion. The utility performs that task, but endpoint logs show it also creates a hidden service and connects to an external command-and-control server. It does not infect other files or spread to other hosts by itself. Which malware type best describes this utility?

- A) Worm
- B) Virus
- C) Trojan horse
- D) Ransomware

QUESTION 4 · EXAM SCENARIO

Several departments report that business documents now have unusual extensions and cannot be opened. A text file appears in each affected folder demanding cryptocurrency payment. File server shadow copies have been deleted, and the encryption process is still running on some hosts. Which malware type is most likely involved?

- A) Rootkit
- B) Ransomware
- C) Fileless malware
- D) Worm

QUESTION 5 · EXAM SCENARIO

A security analyst suspects a compromised Linux server, but standard administrative tools do not show the suspicious process, network connection, or user account. The malware appears to intercept system calls and alter kernel data structures to conceal its activity and persist after reboot. Which malware type best matches this description?

- A) Keylogger
- B) Rootkit
- C) Trojan horse
- D) Polymorphic virus

QUESTION 6 · EXAM SCENARIO

Employees in a call center report slow workstation performance. An investigation finds a background process capturing every keystroke, including login credentials and customer credit card numbers, and periodically sending the collected text to an external server. Which malware type is most directly indicated?

- A) Keylogger
- B) Rootkit
- C) Ransomware
- D) Worm

QUESTION 7 · EXAM SCENARIO

A malware sample is analyzed. Each time it infects a new system, the encrypted payload and part of the decryption routine are different, so signature-based detection fails. After execution, however, the core malicious behavior is the same. Which evasion technique is being used?

- A) Packing
- B) Polymorphic malware
- C) Metamorphic malware
- D) Fileless malware

QUESTION 8 · EXAM SCENARIO

During reverse engineering, analysts observe that each generation of a malware sample has completely different instruction sequences, API calls, and code layout, yet it performs the same credential-stealing function. No significant code block remains identical between samples. Which technique is most likely implemented?

- A) Polymorphic malware
- B) Metamorphic malware
- C) Packing
- D) Rootkit

QUESTION 9 · EXAM SCENARIO

An endpoint detection tool alerts on a Microsoft Word document that spawns PowerShell. The script downloads additional instructions, runs them directly in memory, and creates a WMI event subscription for persistence. Investigators find no malicious executable saved on disk. Which malware category best describes this activity?

- A) Fileless malware
- B) Virus
- C) Worm
- D) Trojan horse

QUESTION 10 · EXAM SCENARIO

A suspicious executable is submitted for analysis. It appears highly compressed, has few readable strings, and when run it unpacks additional code into memory before executing its payload. Signature-based antivirus engines do not detect the original file. Which malware-hiding technique is most likely being used?

- A) Polymorphic malware
- B) Metamorphic malware
- C) Packing
- D) Fileless malware

ANSWER KEY & COMPREHENSIVE DISTRACTOR EXPLANATIONS

Question 1: Option B — Virus

A virus requires a host file and often user action to execute and spread; it infects files such as executables, and other systems are compromised when those infected files are run. A worm self-propagates across the network, a rootkit conceals activity, and ransomware encrypts data for extortion.

Question 2: Option B — Worm

A worm is self-replicating and can spread across networks without user interaction, often exploiting vulnerable services such as SMB. A virus normally needs a host file or user action, a Trojan relies on disguising itself as legitimate software, and a keylogger captures keystrokes rather than self-propagating.

Question 3: Option C — Trojan horse

A Trojan horse disguises itself as legitimate or useful software but performs malicious actions, such as creating a backdoor or contacting command-and-control infrastructure. It does not necessarily self-replicate like a worm or infect files like a virus, and this scenario does not describe encryption for extortion.

Question 4: Option B — Ransomware

Ransomware encrypts files or systems and demands payment, often deleting shadow copies to prevent easy recovery. Rootkits hide activity, fileless malware runs in memory, and worms self-propagate; none of those directly explain the ransom note and mass file encryption.

Question 5: Option B — Rootkit

A rootkit is designed to gain privileged access and hide its presence by modifying kernel structures, system calls, or administrative tools. A keylogger captures input, a Trojan disguises itself as useful software, and a polymorphic virus changes code to evade signatures, but hiding deep system activity is characteristic of a rootkit.

Question 6: Option A — Keylogger

A keylogger records keystrokes to steal sensitive information such as credentials and payment card data. The scenario does not describe file encryption, self-propagation, or deep system concealment, so keylogger is the most direct match.

Question 7: Option B — Polymorphic malware

Polymorphic malware changes its encryption or decryption routine and payload appearance with each infection while keeping the core functionality the same. Metamorphic malware rewrites its entire code, packing hides code through compression or encryption, and fileless malware executes in memory.

Question 8: Option B — Metamorphic malware

Metamorphic malware rewrites its own code entirely, changing instruction sequences and structure while preserving malicious functionality. Polymorphic malware typically mutates an encryption or decryption component, packing compresses or encrypts code, and a rootkit focuses on concealment and privilege.

Question 9: Option A — Fileless malware

Fileless malware uses legitimate system tools, scripts, and memory-resident execution, often leaving little or no executable file on disk. WMI subscriptions can provide persistence. A virus infects files, a worm self-propagates, and a Trojan disguises itself as useful software.

Question 10: Option C — Packing

Packing compresses or encrypts malware code to obscure strings and signatures, then unpacks the real payload into memory at runtime. Polymorphic and metamorphic techniques change code across infections, while fileless malware is characterized by memory-only execution without a dropped executable.